Installation Guide

Summary

Upload the XMIT file `nfsd.v*.xmi` to MVS, to a dataset name of your choice and then "receive" the XMIT data into a PDS, for example `SYSS.NFSD.V0R1M0.DISTRIB`.

This will be a partitioned dataset, `RECFM=FB,LRECL=80` format that contains --

- Install JCL.
- A sample started task JES2 stored procedure.
- A sample configuration which you can place in its own dataset, or a member in `SYS*.PARMLIB`.
- XMIT format data to create the runtime load library.
- XMIT format data for the source datasets.

There are two executable programs which are used by the started task.

- **RESSOCK** - This program is similar to Jason Winter's **RESET** sample program. However, *RESSOCK* takes a list of port numbers for which sockets are to be reset. This program was very useful when NFSD was unstable and would crash leaving socket connections behind it. However, as NFSD (famous last words) is quite stable this probably isn't necessary at this time.
- **NFSD** - This is the executable for the NFS server as you might expect.

Note that the installation job does not copy the configuration file, or started task stored procedure. It is left for you to do this manually.

Note that the full source and development tree can be found on github at --

https://github.com/twinslow/mvs_nfsd

Installation Steps

Step 1 - Load the `nfsd_v*.xmi` file to MVS

Use FTP or your terminal emulator (such as the wonderful Vista TN3270) to load the file to MVS. Then use RECV370 to create the distribution dataset.

Note, that I had to create the dataset first for FTPD and a IND$FILE transfer using Vista TN3270.

The full screen interface provided in MVS-TK5 works (option M.R) and the following inputs, with "option 1" were successful.

```
--------------------   Receive XMIT File   ---------------------------------
OPTION  ===> 1

 1  Receive XMIT file on MVS to PDS/SEQ file                        HERC01
 2  Receive XMIT file on PC  to PDS/SEQ file                        PRECV372
                                                                    PXMI
 XMIT Input File
1 MVSFILE: SYSS.NFSD.V0R1M0.XMI
-or-
2 PCFILE:
  Reader Control for PCFILE
   HercRDR Jes2RDR DEVINIT Reset Command



 MVS Output File
  DSN: SYSS.NFSD.V0R1M0.DISTRIB                     VOL: TSO003  UNIT: SYSDA
  SPACE: (CYL,(5,2,3))              DISP: (NEW,CATLG)

 JOB STATEMENT INFORMATION
  ===> //HERC01N JOB CLASS=A,
  ===> //     MSGLEVEL=1,MSGCLASS=X,NOTIFY=HERC01
  ===>
  ===>
```

In these instructions I am assuming this output dataset is -- `SYSS.NFSD.V0R1M0.DISTRIB`.

Step 2 - Create the NFSD source and load library datasets

Customize the install job `SYSS.NFSD.V0R1M0.DISTRIB(INSTALL)`. You will likely wish to change --

- Job card information such as job name, job class, message class, notify userid etc.
- The dataset name of your `DISTRIB` PDS.
- The target output dataset prefix for load library and source datasets.
- The target volume for these datasets.

Note that these datasets are created as DISP=(NEW,CATLG) and the job will fail if they already exist.

The NFSD server does not need to run authorized, but it will need to have appropriate permissions to its configuration and the datasets to be exported (the datasets that are to be remotely mounted and accessed).

In the *MVS TK5* environment, the started task will run under the `STC` userid, and have a group of `STCGROUP`. The permissions in the default setup should be suitable, allowing read/write access to any dataset you wish to export. See *TK5* and *RAKF* documentation for more information on this.

Step 3 - Customize, or create the started task stored procedure

The member `SYSS.NFSD.V0R1M0.DISTRIB(NFSD)` is a sample stored procedure you can copy to a suitable JES2 procedure library. If you are using the MVS TK5 system then I suggest using `SYS2.PROCLIB`.

- Update the load library dataset as required.
- Uncomment and change SYSOUT class, or destination of SYSUDUMP as appropriate.
- Update the source of the configuration file (see below).

Step 4 - Customize, or create the NFSD configuration file

The member `SYSS.NFSD.V0R1M0.DISTRIB(CONFIG)` contains a sample configuration for the NFSD server. You can copy this to either a sequential dataset, or a member of a PDS. The dataset can be record format FB or VB. You way wish to keep the the configuration as a member of `SYS1.PARMLIB`, such as `NFSDCFG0`.

The sample JES2 stored procedure assume exactly this setup.

Note that at this time the dataset names in the export list cannot include wilcards to perform generic exports based on system catalog. You must explicitly list each dataset you wish to export.

Step 5 - Start NFSD

Enter a start command on the MVS console to run NFSD, `S NFSD`.

MVS Modify Commands

The logging level and options can be changed via a MVS modify command.

For example to set the level of messages being output to the log use the command --

```
F NFSD,SET LOGLVL INFO
```

The log levels (INFO is show above) are --

- DEBUG
- TRACE
- INFO
- WARN
- ERROR
- FATAL

You can separately set the level of messages that are output as WTO messages to the console.

```
F NFSD,SET WTOLVL WARN
```

The above command will cause NFSD to output only WARN messages and worse, to the MVS console.

You can also independently set logging level for different NFS operations.

```
F NFSD,SET LOGLVL DEBUG PROC=WRITE
```

This would cause any NFS WRITE operation to output DEBUG level messages, overriding (for example) a global INFO logging level.

The NFS operations that can be logged using the `PROC=` keyword are --

- GETATTR
- SETATTR
- LOOKUP
- ACCESS
- READ
- WRITE
- CREATE
- REMOVE
- RENAME
- READDIR
- READDIRPLUS (or the alias of RDIRPLUS)

- FSSTAT
- FSINFO
- PATHCONF
- COMMIT
- NULL

Shutting down NFSD

Issue an MVS STOP command to shutdown NFSD --

```
P NFSD
```